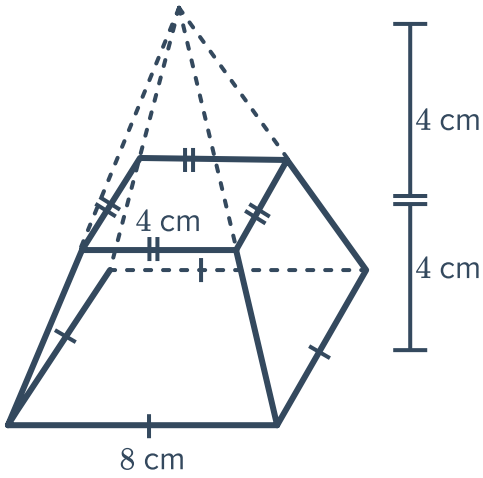
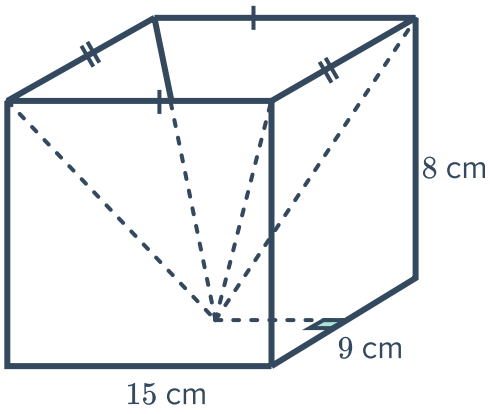
Volume of truncated pyramids and cones

1 A small square pyramid of height 4 cm was removed from the top of a large square pyramid of height 8 cm forming the solid shown.

Find the exact volume of the solid.

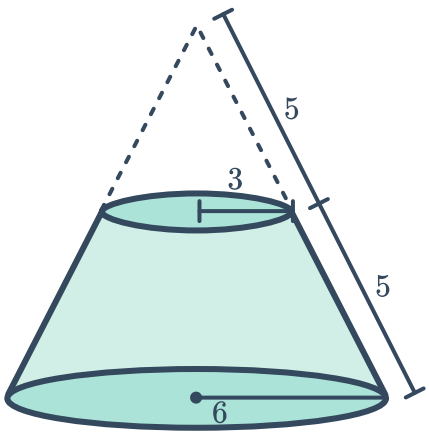


2 A pyramid has been removed from a rectangular prism, as shown. Find the volume of the remaining solid.



3 The top of a solid cone was sawed off to form the solid attached, such that radius of the top face is 3 cm and the radius of the bottom face is 6 cm.

Find the volume of the solid formed. Round your answer to two decimal places.



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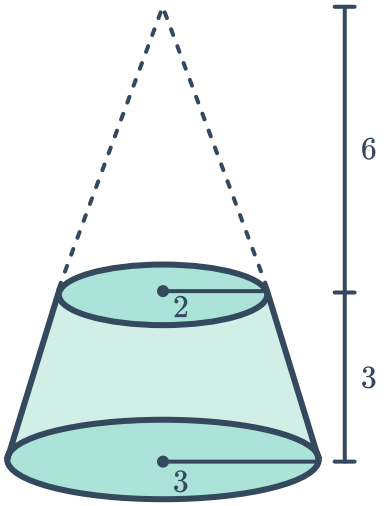
4 Each of the following solid is a truncated cone, formed by cutting off a cone shaped section from the top of a larger, original cone.

i Find the exact volume of the cone section that was cut off.

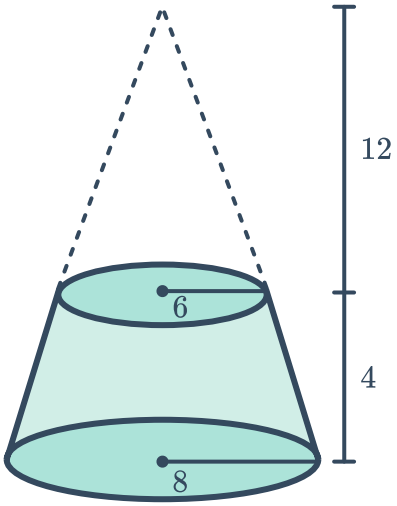
ii Find the exact volume of the original cone.

iii Hence, find the volume of the truncated cone, correct to two decimal places.

a



b

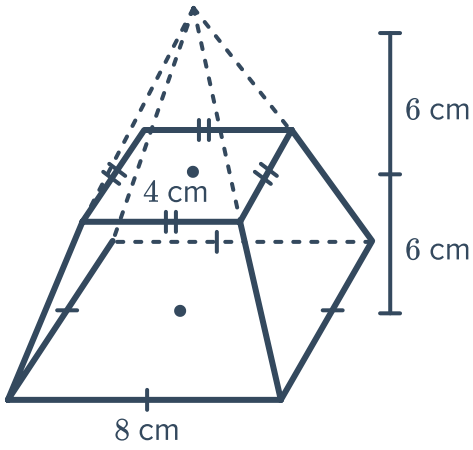


Surface area of truncated pyramids and cones

5 A small square pyramid of height 6 cm was removed from the top of a large square pyramid of height 12 cm to form the solid shown.

a Find the length of the slant height of the sides of the new solid, correct to two decimal places.

b Hence, find the surface area of the solid formed, correct to one decimal place.



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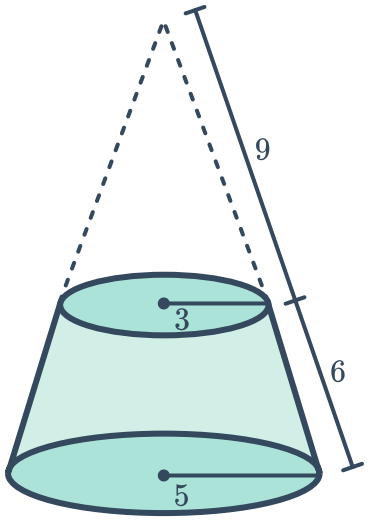
6 Each of the following solid is a truncated cone, formed by cutting off a cone shaped section from the top of a larger, original cone.

i Find the exact curved surface area of the original cone.

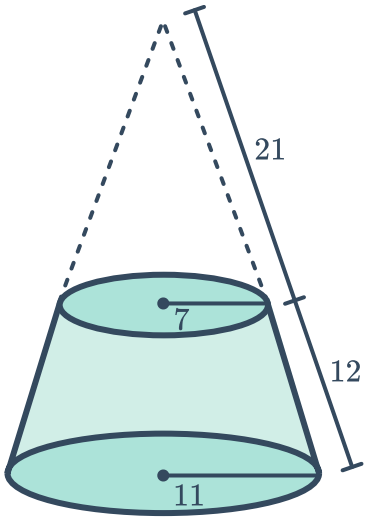
ii Find the exact curved surface area of the cone section that was cut off.

iii Hence, find the surface area of the truncated cone, correct to two decimal places.

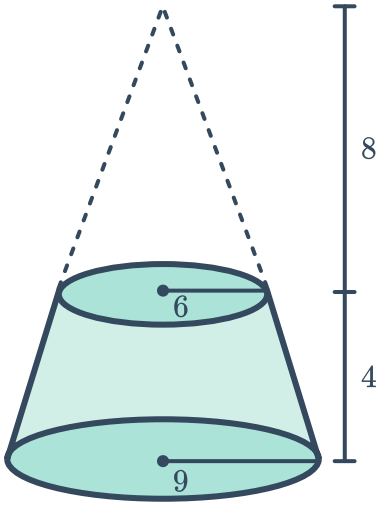
a



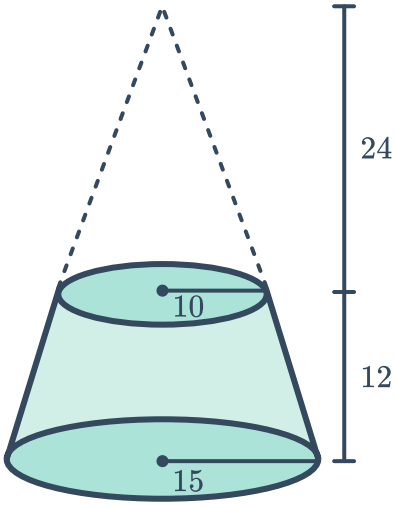
b



c



d

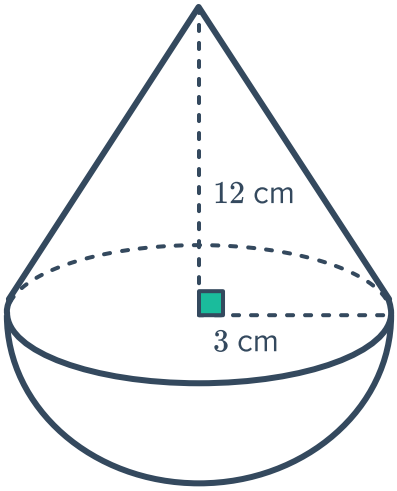


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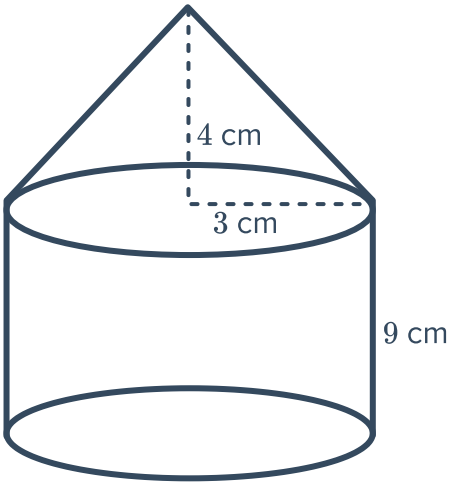
Other compound solids

7 Find the volume of the solids correct to two decimal places:

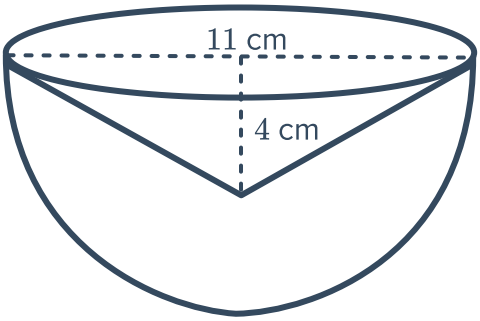
a



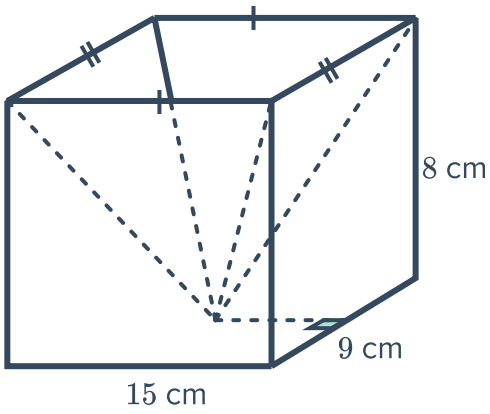
b



c



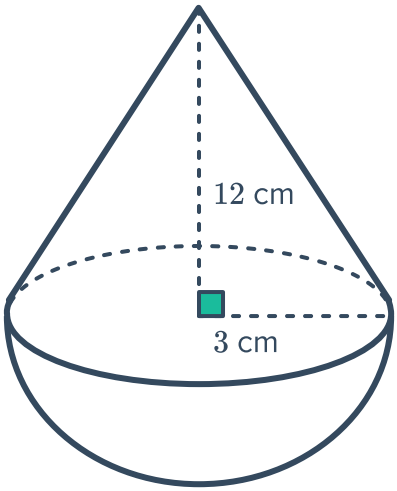
d



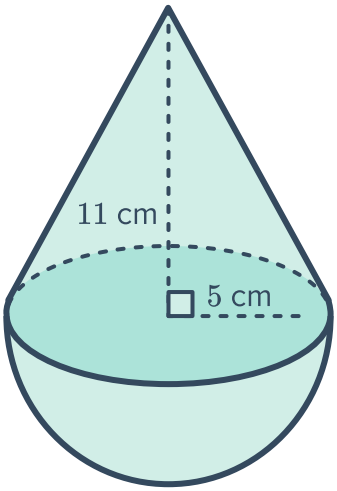
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8 Find the surface area of the following compound figures, correct to two decimal places:

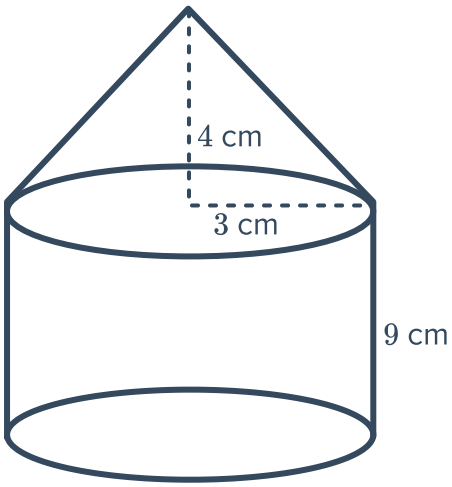
a



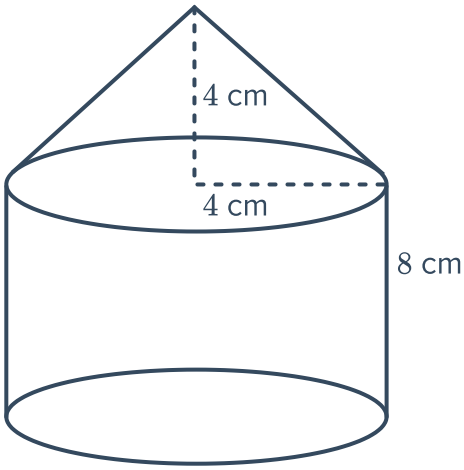
b



c

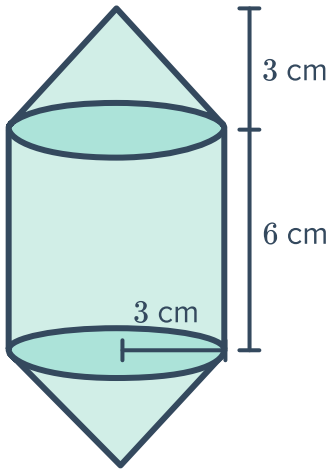


d



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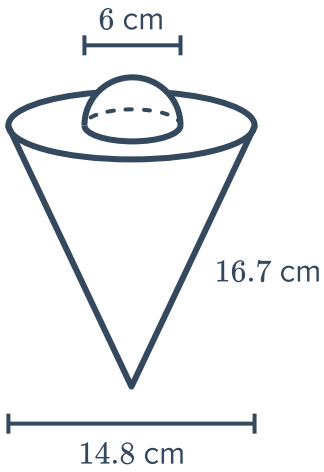
9 The figure shows a cylinder of radius $3\ \text{cm}$, and its height is double the radius. On the top and bottom of the cylinder are cones with radii and height both also equal to $3\ \text{cm}$.



a Describe the steps involved in calculating the surface area of this compound solid?

b Find the surface area of the solid. Round your answer to two decimal places.

10 Consider the following solid:



a Find the perpendicular height of the cone, correct to one decimal place.

b Find the volume of the solid, correct to two decimal places.

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Applications

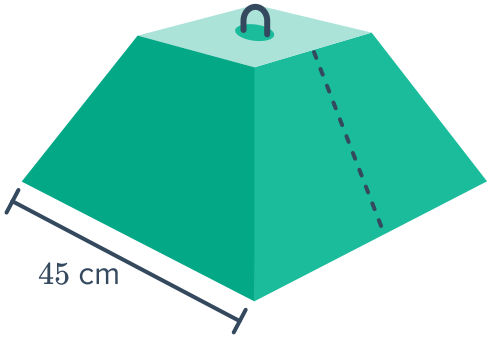
11 A weight is constructed by removing the top $38\ \text{cm}$ from a $57\ \text{cm}$ tall pyramid with a square base of side length $45\ \text{cm}$.

a Find the volume of the original pyramid.

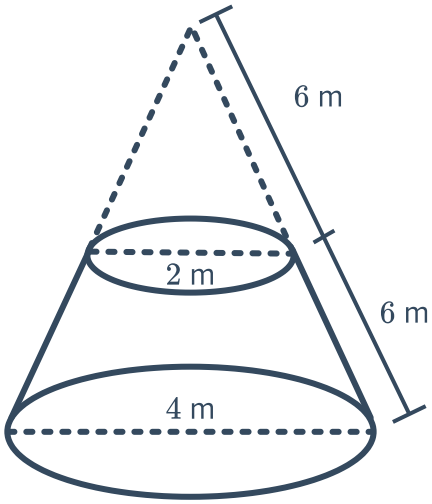
b Calculate the side length of the square on top of the weight.

c Calculate the volume of the removed part of the pyramid.

d Calculate the volume of the weight.



12 The following podium was formed by sawing off the top of a cone. Find its volume, correct to two decimal places.



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13 Before 1980, Mount St. Helens was a volcano approximately in the shape of the top cone below:

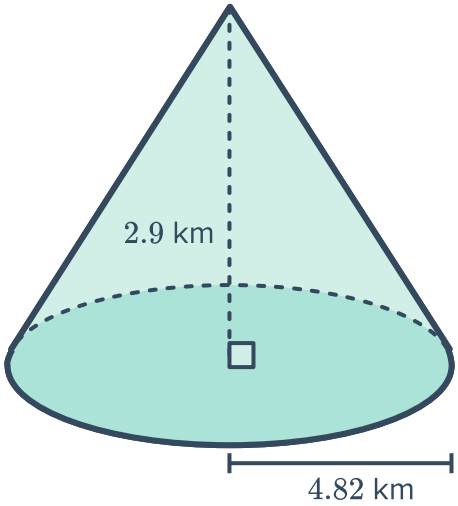
a What was the volume of the mountain, in cubic kilometres? Round your answer to two decimal places.

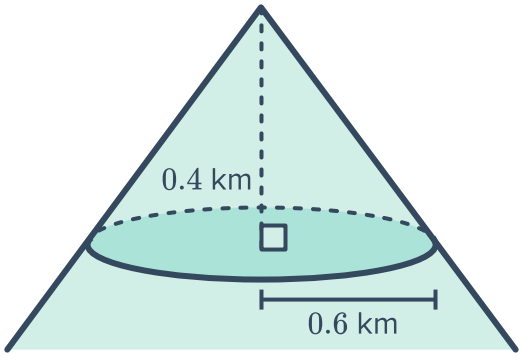
b The tip of the mountain was in the shape of the bottom cone shown.

Find the volume of the tip in cubic kilometres. Round your answer to two decimal places.

c In 1980, Mount St. Helens erupted and the tip was destroyed.

Find the volume of the remaining mountain, in cubic kilometres. Round your answer to two decimal places.





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